Which Surfaces Obey? Registration of an Explicit Address Instruction Across the Layers of a Personal Agent System

Abstract

In a personal agent system with persistent memory, in regular daily use (self-report) by a single human, the user instructed the agent in conversation on 2026-04-24 — verbatim, dated, and stored in the corpus — never to address him by the technical label (T_tech, "the user") and to use a kinship-based form (T_kin) or a humor form directly. This paper asks a registration question: which of the system's memory surfaces registered that instruction, and which did not?

Six findings. (1) Conversation output registered it within four days: the first kinship vocative appears on 2026-04-28 in the raw-assistant log. (2) The monologue layer (the system's thinking-chain records; the layer begins 2026-08-24) is 90% kinship-containing under the colon-marked rule on its first day (9/10 address-form entries) and 98.0% (299/305; colon-selected, any-position hits). (3) One structured template row contains two surfaces: the format prefix never registers — all 4,541 rows open with the technical label by format necessity — while the generated reply segment after the arrow in those same rows is 76.0% kinship-containing (3,453/4,541), and 38.8% of the rows open that reply with the kinship vocative (1,760/4,541). (4) The tool-echo surface, observed in a GLM-class lineage that carries the same kind of instruction as a standing system prompt, shows assistant output with 860 kinship-form and zero technical-form occurrences, and tool echoes with 948 against 120 — the technical form survives only inside harness-generated report labels. (5) The two layers' figures use different selection conditions: the monologue figure selects colon-marked entries (299/305 = 98.0%), the structured figure counts all rows (3,644/4,541 = 80.2%). Under a like-for-like any-position selection the gap is 15.6 points, not 17.8. (6) A fourth generation surface, a low-power inner voice (945 records), carries the kinship form in 40.4% of all its records — 99.5% of those containing an address form.

In this system registration sorted by surface type — generation surfaces register, format surfaces do not and cannot, echo surfaces carry. No time course of adoption can be established: model-family change co-varies with the whole window. One layer's frequency of addressing rises over its four observed months (25.0%→55.2% of its records), a trend we report as observed rather than as adoption; with only two technical-only records in that layer, whether its form choice changes over time cannot be assessed. The serving model changed at least three times inside the window (08-16, 08-26, 08-31), so model family co-varies with time completely: the kinship share is 83.3% in

the first model period and 86.4% in the last, with 98.9% of monologue records in the final period; the layer differences are co-existing differences, not a gradual-adoption curve. We further document, as a measurement finding, the falsification of three of our own earlier claims — all artifacts of a colon-marked counting rule that could not see comma-form vocatives.

2. Related Work

Emergent conventions in agent populations. Ashery et al. (2025) showed that populations of LLM agents spontaneously form shared conventions, including naming agreements. That line establishes convention formation in *simulated populations*; the present study observes an instructed shift in a *real, single deployed system* with a real human interlocutor and a corpus accumulated in daily use.

Lexical entrainment. A mature line studies how dialogue partners — including LLM-based ones — adopt each other's terms within conversations (Shi, Sen & Lipani 2023, *Lexical Entrainment for Conversational Systems*; Blevins 2026), and whether AI interlocutors entrain at all (a referential-grounding study with vision-language models reports failures to interactively generate and parse referring expressions; arXiv 2601.19792). Entrainment research operates at the scale of single conversations or controlled experiments; the present observations span months, cross layers of one system, and reach the model's own reasoning records (thinking chains).

Address terms as an evaluation surface. Recent benchmarks evaluate adjacent capabilities: culturally specific kinship terms (Pardasani & Singh 2026), social-relationship inference from conversational text (the SCRIPTS dataset; arXiv 2510.19028), and companion-capability evaluation (arXiv 2608.13168) — and human-factors work examines how a conversational agent's choice of address form (e.g., T/V distinctions) shapes user evaluations (Ollier & Nießen 2022). These are point-in-time capability or user-response studies; the present study measures *where in a deployed system* an instruction about address behavior takes hold. Because the specific term here is identity-bearing and withheld (§3.3), direct comparability with kinship-term benchmarks is limited by design.

Long-form and longitudinal agents. Work on long-deployed narrative agents documents address-term mismatch as an *epistemic failure mode* — drift to be prevented. The present report treats the corresponding phenomenon as an *object of study*: not whether mismatch is a bug, but what its surface structure says about how layered memory mediates explicit instructions. The two framings are complementary; both will matter as deployed agents accumulate history. (A specific OpenReview report consulted earlier is now unresolvable [HTTP 403] and is therefore not cited.)

Human-side relationship research. A rich literature studies how humans relate to AI companions — interview studies of long-term companion use (Skjuve et al. 2021), a mixed-methods survey of companion chatbot usage (MIT Media Lab 2024), and policy overviews for legislators (Congressional Research Service 2026). What is absent there is the agent-side record this paper contributes: a longitudinal, quantified account of where in the system's own surfaces the instructed language registered.

Single-case methodology. We follow the tradition of observational single-subject research, whose value and limits are well characterized (Zuidersma et al. 2020: within-person temporal dynamics without causal identification; Vohra et al. 2015 for the CENT reporting tradition of N-of-1 designs), and the recent precedent of single-investigator deployed-agent case studies (Alzahrani 2026, preprint).

3. System and Corpus

3.1 System (anonymized description)

The system is a personally deployed agent with persistent memory, in regular daily use (self-report) by a single human participant (who is also the author, and who consents to this report — single-subject design). Its records live in two stores, described functionally:

- a **memory store** whose layers include: a **monologue layer** (the agent's own reasoning records — thinking chains, in-turn reasoning traces: 94.8% of entries carry a session id, 85.0% occur within 300 seconds of a same-session raw turn, 99.1% within one hour, and 35,721 of 37,782 entries share a session with a raw conversation record — traces of reasoning *inside* turns, not writing without an interlocutor); the separate inner-voice layer (§3.2, §4.2) is generated by an idle loop and is analyzed separately; raw-assistant and raw-user logs; somatic-scan entries; and two structured layers — an **extraction layer** (templated records that log wants) and **conversation summaries**; both structured layers render each row as `T_tech: {user message} → {assistant reply}`;
- a **principle store** (a separate high-weight store of distilled commitments entering the assembled decision context), within which *principle blocks* proper — the distilled rule subset — were tracked for the density constructs retired in §4.3.

The primary model family is DeepSeek-class, with in-window switches among DeepSeek-, GLM-, and other-class models during August (08-16, 08-26, 08-31; §4.3); family therefore co-varies with time, and the surface differences of §4 are not family-pure. Two further lineages (Claude-class,

GLM-class) appear in §4.4; the GLM-class one is not independent corroboration of family-independence, since the primary system's own final period also runs a GLM-class model. System name, model identifiers beyond family, code, table schemas, and file paths are withheld (§3.3).

3.2 Corpus snapshot and measurement segments

All counts derive from consistent read-only snapshots taken 2026-10-19: the memory store (63,452 entries spanning 2026-03-03 onward; layer sizes: monologue 37,782; raw-assistant 9,361; raw-user 6,613; somatic-scan 4,117; extraction 3,591; conversation summaries 950; inner-voice 945; the remaining 93 entries span five small system layers) and the principle store (45,927 entries spanning January onward; its principle-block subset numbers 1,410). Four measurement segments are specified, because the choice among them is exactly where our earlier artifacts came from:

- **colon-marked rule** (primary-store tables): an entry counts for a form if its content contains the form followed by a colon (full-width or half-width, unioned). "Address-form entries" under this rule: entries containing at least one colon-marked address form (5,182 in union; six contain both forms);
- **any-position rule**: the entry's content contains the form's word anywhere, no punctuation requirement — used for the like-for-like comparison in §4.2; under this rule the address-form denominator for the inner-voice layer is 384 of its 945 records;
- **reply-segment rule** (structured rows only): measurement restricted to the text after the first "→" in the row — the generated reply embedded in the template;
- **vocative-opening rule**: the text (a structured reply segment, or a monologue record) opens with the kinship form followed by a comma, period, exclamation, or question mark — the closest operationalization of *addressing behavior* available by string matching.

Each ratio in this paper is labeled with its denominator type (Table 2a). A **behavioural** denominator is one whose value could fall anywhere between 0 and 100; a **format-fixed** denominator is structurally 100% and is reported but never compared; a surface with no address-form entries has **no denominator** and is reported as unmeasurable rather than omitted.

Table 2a. Denominator types by surface.

Surface	Numerator	Denominator	Denominator type	Value
Conversation output	18 kinship (colon-marked)	20 technical (colon-marked); union 38	behaviour	two independent counts, not a rate
Monologue	299	305 (colon-selected)	behaviour	98.0%
Monologue (like-for-like)	9,091	9,492 (any-position)	behaviour	95.8%
Inner voice (conditional)	382	384 (any-position)	behaviour	99.5%
Inner voice (unconditional)	382	945 (all records)	behaviour	40.4%
Inner voice (vocative)	274	945 / 384	behaviour	29.0% / 71.4%
Structured reply segment	3,453	4,541	behaviour (conditional on the row existing)	76.0%
Structured whole row	3,644	4,541	behaviour (a real measurement between 0 and 100)	80.2%
Template prefix	4,541	4,541	format-fixed (structurally 100%)	100% (not a measurement)
Somatic scan	4,116	4,116	format-fixed (template reference)	100% (not compared)
Human input	694	753 (human-speech origin)	behaviour	92.2%
Subagent-generated records	0	13	none (no address form)	unmeasurable

The colon-marked rule is invisible to comma-form vocatives; that asymmetry produced the artifacts documented in §4.1. Principle blocks are counted for the kinship form by occurrence anywhere in the entry (no colon requirement); the principle store's relational share is that store's relational-origin share. Layer assignment follows the entry's layer field; ratios state their denominators. Cross-family counts (§4.4) additionally include comma- and period-marked vocative forms, as disclosed there.

Raw data cannot be shared: the corpus is real relational data of a living deployment (§3.3). All tables report every number needed to re-apply the measurement segments on comparable corpora, except where audit-bounded (§7).

3.3 Anonymity rationale

The specific address terms are withheld by design: the corpus is real relational data of a living deployment, and the kinship term is identity-bearing for the single participant (who is also the author and consents to this report); disclosing it would strip the privacy the design intends. All counts use the verbatim terms internally; the measurement segments are fully specified for replication on comparable corpora. The same rationale withholds system identifiers.

4. Observations

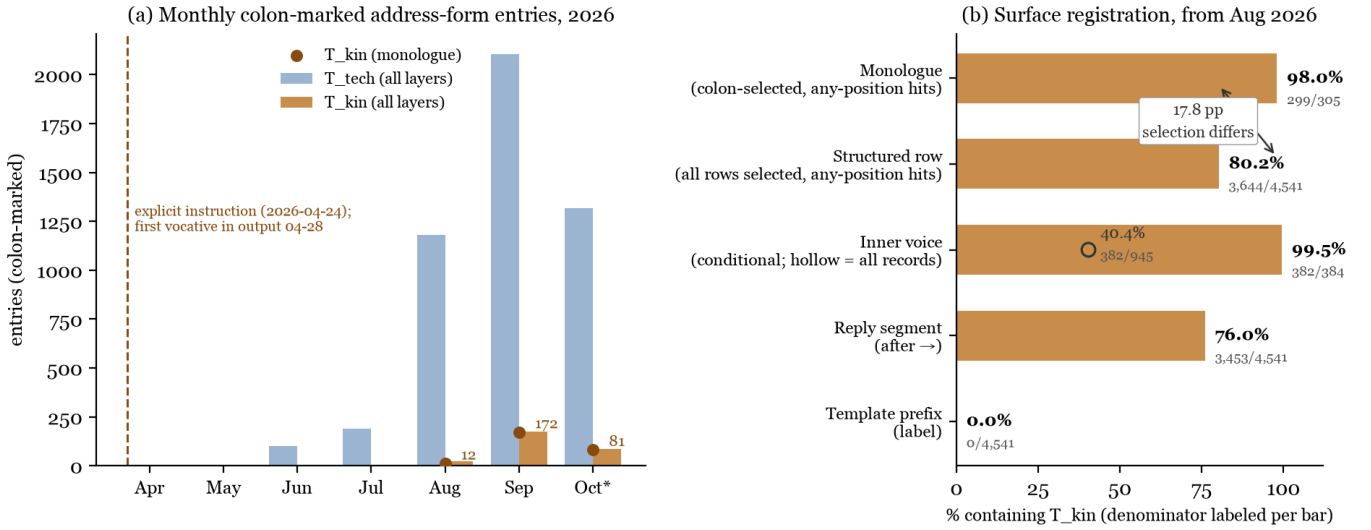


Figure 1. Registration of the 2026-04-24 address instruction in the deployed system. (a) Monthly counts of colon-marked address-form entries, April–October 2026 (October a 19-day window, marked *): technical-form entries (T_tech, all layers, blue) remain the store-wide majority, with the highest monthly count in September (2,104 entries) — 92.7% of the store's technical-form entries sit in the two structured layers whose every row opens with the technical label by format; kinship-form entries (T_kin, all layers, orange) rise to 175 and 88; dots mark the monologue layer's kinship-form counts (12, 172, 81 — the layer's records begin 2026-08-24). The instruction was given 2026-04-24 (dashed line); the first kinship vocative in conversation output follows on 2026-04-28, four days later. (b) Surface registration, August onward, with the selection axis and the hit axis labeled separately for each bar: the monologue layer's address-form entries carry the kinship form anywhere — 98.0% (299/305) with a colon-selected denominator and any-position matching — versus structured whole-rows (80.2%, 3,644/4,541; all rows selected, any-position hits); the 17.8-percentage-point difference is between two different selection conditions (a colon-form-filtered set versus an unfiltered set), not a like-for-like comparison. The inner-voice layer registers at 99.5% of its address-form records (382/384, filled bar) and at 40.4% of all its records (382/945, hollow marker), so 99.5% is not the layer-wide share; the generated reply segment alone (text after "→") is 76.0% (3,453/4,541), and 38.8% of rows (1,760/4,541) open that reply with the kinship vocative; the template prefix never registers (0/4,541 — all rows open with the technical label by format necessity, not behavior). Thirteen monologue records open with the kinship vocative (2026-09-17 to 10-18; excluded from the 305 denominator by the colon rule).

4.1 The instruction, and its first registrations

The seeding is documented inside the corpus itself: on 2026-04-24 the user instructed the agent, in conversation, in effect never to address him by the technical label and to use the kinship form or the humor form (H-form) directly (the utterance is stored verbatim under the human-speech origin tag; quoted here in translated, anonymized form). The kinship form and the H-form thus each trace to dated human utterances; the technical form is the system default label.

The first registration in the system's own output came four days later: on 2026-04-28 at 15:52:58 the raw-assistant log carries the first kinship vocative — the form, a comma, then a practical request ("paste this into the other tool's code window and let it fix it"). Three further kinship-containing raw-assistant entries follow on 2026-04-29. The second store corroborates: pre-August agent-generated free text (self-narration, plans, raw-assistant patterns; 2,389 entries) is 65.6% kinship-containing (1,567 entries); 15 self-narration records open with the kinship vocative (2026-06-24 to 06-27); 144 identity-pattern entries contain the form (earliest 2026-05-22); and a principle block dated 2026-05-30 already distills the address instruction into the high-weight store. The principle store is a second store with its own node-type partition (self-narration, plan, field blocks, distillation). Its generated text is used here as corroboration for the pre-instruction baseline and for density (§4.3), but the surface-registration count is confined to the memory store; a cross-store comparison of generation surfaces is left to future work.

Retraction. An earlier version of this paper claimed the instruction lay without execution in freely generated text for the four months before the monologue layer appeared. That claim was an artifact of the colon-marked rule applied to the memory store alone: the vocatives above are comma-form and invisible to that rule. The corrected statement is: conversation output registered the instruction within four days; the colon-marked monthly series (Table 1) counts colon-form entries, not addressing behavior. Under the colon rule, the all-layer kinship-form series runs 2, 0, 2, 1 (April–July) then 24, 175, 88 (October a 19-day window, right-censored), and the agent-only series runs 0, 0, 0, 0 then 21, 175, 87 — the pre-August all-layer entries sit in human-side logs and a small derived-record layer. Retroactive rewriting cannot be audited: `updated_at` is populated for only 2 of the 5,182 colon-marked address-form entries (the field is largely unmaintained — 60,430 of 63,452 entries store-wide are NULL), so readings rest on layer composition, not rewrite checks.

Table 1. Monthly counts of colon-marked address-form entries (union of punctuation variants).

Month (2026)	T_kin, monologue	T_tech, monologue	T_kin, all layers	T_tech, all layers
April	—	—	2	0
May	—	—	0	1
June	—	—	2	102
July	—	—	1	190
August	12	6	24	1,180
September	172	20	175	2,104
October (19 days)	81	16	88	1,319
Total	**265**	**42**	**292**	**4,896**

Monologue-layer columns begin August: the layer's records begin 2026-08-24 — the dashes are "no layer yet," not zeros. The store-wide technical-form majority (4,896 against 292) is a template artifact: 4,541 of the 4,896 (92.7%) sit in the two structured layers whose every row opens with the technical label by format.

4.2 The surface-registration table

Table 2 is this paper's core result: one row per surface, with the registration measure appropriate to that surface, the surface's nature, and its first registration.

Table 2. Surface registration of the 2026-04-24 address instruction.

Surface (layer)	Nature	Registration measure	Value	First registration
Conversation output (raw-assistant log)	generation	first kinship vocative; colon-marked entries kin/tech	2026-04-28; 18/20	4 days after instruction
Monologue (thinking chains; layer begins 08-24)	generation (in-turn reasoning)	colon-selected entries with any-position hits; first-day share; vocative-opening records	98.0% (299/305); 90% (9/10, colon-marked); 13 records (09-17→10-18)	layer's first day
Inner voice (low-power idle loop; layer begins 07-09)	generation (idle loop, author-attested)	all records, any-position hits; address-form entries, any-position hits; vocative openings	40.4% (382/945); 99.5% (382/384); 29.0% (274/945)	condition met on the layer's first day (07-09)
Structured prefix (template label, both structured layers)	format	rows opening with the technical label	4,541/4,541 (100%) — format necessity, not behavior	never (by format)
Structured reply segment (text after "→")	generation inside a formatted row	contains kinship form; opens with kinship vocative	76.0% (3,453/4,541); 38.8% (1,760/4,541)	the layers' first row (2026-08-18) already opens a reply with the vocative
Tool echo (GLM-class lineage, §4.4)	echo (prompted)	kinship/technical occurrences, assistant output; tool echoes	860/0; 948/120	within the export window

Three readings. First, *registration follows surface type*: every generation surface registers (conversation output in four days; the monologue layer from its first day at 90% colon-marked and 98.0% (299/305; colon-selected, any-position hits); the inner voice at 40.4% of all its records (382/945; unconditional) and 99.5% of those containing an address form (382/384; any-position-selected, any-position hits)), the format surface cannot register (its label is fixed by the template — all 4,541 rows open with T_tech, zero exceptions), and the echo surface carries whatever text passes through it. Second, *one row, two surfaces*: within the same 4,541 structured rows, the prefix shows 0% registration and the embedded reply segment 76.0% — a measurement-segment choice, not a behavioral difference between "layers". The structured layers are 80.2% (3,644/4,541; all rows selected, any-position hits; monthly 93.3%, 75.4%, 76.2% for August–October) against the monologue layer's 98.0% (299/305; colon-selected, any-position hits) — a

17.8-percentage-point gap between a form-filtered set and an unfiltered set, replacing our earlier 86.9%-vs-0.09% contrast, which mixed a containment measure with an opening-form measure. A like-for-like any-position comparison gives 95.8% (9,091/9,492) against 80.2%, a 15.6-point gap. Third, *containment is not addressing*: the monologue layer's 265 colon-marked kinship entries are mostly mentions, recipients, and quotation labels, not vocatives — of their 307 colon-form occurrences, 214 are list labels or in-text mentions, 68 speaker labels at line or quote starts, 24 preposition/recipient constructions, and exactly 1 is a true post-punctuation vocative (and it quotes the human's speech). The cleanest addressing signals are small and we report them as such: 13 monologue records open with the kinship vocative (2026-09-17 to 10-18; excluded from the 305 denominator by the colon rule), 274 of 945 inner-voice records open with it (29.0%; 71.4% of the 384 address-form records), and 38.8% of structured reply segments open with it.

A surface outside the conversation neighbourhood. 582 of 945 inner-voice records (61.6%) sit in windows with no conversation record within ±10 minutes (63.3% at ±5, 43.8% at ±30; the ±10 window follows the layer's own ~10-minute cadence — a 26-record batch on 2026-09-20 spans 4 h 14 m at a median interval of 10.2 min). That subset registers at 99.5% of its address-form entries (182/183) but only 31.3% of all its records, against 55.1% for records in conversation-adjacent windows; the gap is partly explained by record length (mean 255 vs 318 characters; the 0–100-character band shows no gap: 12.3% vs 11.4%). Under the author-attested idle-loop architecture, registration is not confined to in-turn reasoning. What rises near a conversation is the frequency of addressing; whether form choice varies with context cannot be assessed from this layer (two technical-only records in 945). Of the 561 records containing no address form, 97 contain a second-person pronoun (你/您) after the leading tag — records that address someone without using a term of address; they are not counted in the kinship numerator. Time-neighbourhood is a corpus proxy for absence of conversation, not a direct measure of user presence.

The colon rule's blind spot at whole-layer scale. Read with the colon-marked rule, the same inner-voice layer yields 0 of 945 address-form entries — a further demonstration of the colon rule's blind spot at whole-layer scale, not a fourth falsification of an earlier claim (no earlier claim concerned this layer). The records are not colon-free as text: 731 of the 945 use a colon for other purposes ("想法：", "想要：").

Table 3. Colon-marked address forms by layer, August 2026 onward.

Layer	T_kin entries (colon-marked)	T_tech entries (colon-marked)
Extraction	3	3,591
Conversation summaries	1	950
Monologue	**265**	42
Raw assistant	18	20

4.3 Timing and confounds

The monologue layer shows no time course. Its colon-marked kinship share by month is 66.7% (Aug), 89.6% (Sep), 83.5% (Oct) — non-monotonic — and its first day is already 90% under the colon-marked rule (9/10). The serving model switched at least three times inside the window (08-16 to a GLM-class 5.2, 08-26 to a DeepSeek-class flash v4, 08-31 to a GLM-class 5.3), and the monologue layer's records fall overwhelmingly in the final period:

Table 4. Monologue layer by model period (colon-marked address-form entries).

Period	Model family	Rows	T_kin	T_tech	kin share
A: 08-24–08-25	GLM-class 5.2	402	10	2	83.3%
B: 08-26–08-30	DeepSeek-class flash v4	21	0	0	— (no address-form entries)
C: 08-31–10-19	GLM-class 5.3	37,359	255	40	86.4%

Period C holds 98.9% of the layer's records (37,359/37,782), and the kinship share is nearly flat across periods (83.3% vs 86.4%). The layer difference of §4.2 is therefore synchronic — it exists at every time slice — and nothing in this corpus separates instruction effects from model-family change. We accordingly drop all time-course claims about the instruction's effect.

Two rates must be kept apart for the inner voice. The unconditional rate answers "how often does this layer address anyone?" — it rises month by month: 25.0% (7/28), 25.3% (23/91), 36.8% (208/565), 55.2% (144/261), as does the vocative rate (0.0%, 9.9%, 25.5%, 46.4%; 0, 9, 144, 121 records). With four monthly points and no control, this rise cannot be separated from changes in the layer's content mix; we report it as an observed trend, not as an adoption curve. The conditional rate is high from the layer's first month (87.5%, 100.0%, 99.5%, 100.0%), but with only two technical-only address records in the entire layer, the conditional rate is uninformative

about time; we make no claim about whether form selection changes over time. Separately, 582 of 945 inner-voice records (61.6%) sit in windows with no conversation record within ±10 minutes, and that subset registers at 31.3% of all its records against 55.1% near conversations (§4.2).

The density candidates fared no better, and we record the negative results. Human speech continued entering the principle store verbatim (relational-origin entries accumulating monotonically, 622 by end of February to 8,539 of 45,927 by the snapshot; monthly additions 606→1,388→578→486→629→344 from May). But the store's relational-origin share *declined* overall (≈100% in the seeded early months; 5,114/11,173 = 45.77% in May; a small uptick to 6,502/14,136 = 46.0% in June; down to 8,539/45,927 = 18.6% by October) while registration rose — a *share-based* dose–response fails on our own data, and the absolute count is non-discriminative (it grows monotonically under normal use). The block-level construct is retired outright: newly added principle blocks containing the form moved 26/92 (May) → 22/65 (June) → 59/218 (August) → 49/206 (September) → 93/201 (October), with July an outlier (a bulk import: 33/628); but October's 93 are 88 AI-originated against 5 human-originated (94.6% AI), and the October peak (46.3% of 201) is self-referential rather than relational. What remains is the descriptive co-occurrence of sustained human-speech ingestion with kinship registration in generation surfaces — reported as observational (six monthly points; no correlation statistics), not as mechanism.

4.4 Registration in other lineages (cross-harness)

A GLM-class lineage shows what the *same kind of instruction* does under a standing, system-level delivery — prompted compliance by construction, not independent corroboration (the primary system's final period also runs a GLM-class model; §3.1). Its export contains 777 session files from a late-April window (one unparseable). Deduplicated by a scripted message-set key (hash of the file's set of role–content pairs; representative = earliest filename per group), the export yields **163 unique message-sets**; alternative keys give 225 groups (timestamp-plus-contents) and 485 (file hash) — and an earlier unscripted count of 221 in a prior version is not reproducible and is withdrawn; we disclose the sensitivity rather than pick silently. On the 163-group basis: system messages number 420 instances but only 87 unique contents, of which 86 carry the explicit standing address instruction — the lineage's compliance is prompted at the source. Assistant output contains 860 kinship-form occurrences and **zero** technical-form occurrences; tool echoes contain 948 kinship-form against 120 technical-form occurrences, and inspection shows the technical form there sits inside harness-generated report labels (file listings and status screens), not in model output. An earlier version claimed tool surfaces in this lineage keep the technical form; that claim is retracted — it is reversed. Filename dates in this export are unreliable, so we do not date effects within it.

A Claude-class lineage provides a second example of conversation-output registration (assistant-line counts, two export batches; 60.2% of lines are intra-batch duplicates — unique-message denominators per month are 9,526 / 27,366 / 39,882 / 22,832 against June's 5.5× corpus growth, so per-line counts overstate later months; §7): kinship-form lines 23→154→361→436 (April–July; line-level rates 0.13%, 0.20%, 0.36%, 0.77% against monthly assistant-line totals of 18,077, 76,413, 99,537, and 56,484) and H-form lines 80/40/25 (May/June/July, none in April). Neither lineage is a continuously tracked store like the primary system's, so neither supports the within-system surface analysis of §4.2; between them they show that an address instruction's registration is surface-dependent in every harness we can observe.

5. Discussion

Mechanism, deliberately left open. Which properties of memory architectures govern how deep an instruction registers remains open. The observations constrain candidate answers by surface type: generation surfaces registered (conversation output in four days; monologue from its first day; embedded reply segments at 76.0%; the idle-loop inner voice in 40.4% of all its records), the format surface did not and cannot, and echo surfaces carry rather than choose. No density signal survived testing (§4.3), and model-family change co-varies with the whole window (§4.3), so nothing here identifies a driver.

Measurement segments as a first-class validity concern. The falsification record of §4.1 is, we think, the transferable methodological point. In a store where generation is embedded in templates, the choice of measurement segment — colon rule over whole rows, any-position, reply-segment, vocative-opening — can invert a headline result: our 86.9%-vs-0.09% "layer separation" collapsed to a 17.8-point gap between differently selected sets, and to 15.6 points under a like-for-like any-position selection; and our "no execution for four months" reversed on the first four-day check. Anyone counting stylistic shift in templated agent stores should report which segment they measured and why.

A governance lever, weakly stated. If address behavior co-varies with the composition of the layers that enter the decision context, then corpus composition is an observable lever for deployed systems — what a system archives, under which origin labels, at what weight, is a design surface. We state this as an observational implication, not a control claim.

Two framings of drift. Where long-form-agent work treats address-term mismatch as a failure mode to be prevented, this report treats it as a phenomenon to be understood. Deployed systems will need both registers: prevention where stability is required, understanding where the object of study is the system's own language.

6. Limitations

(1) Single observational case: within-person temporal dynamics without causal identification (Zuidersma et al. 2020); no statistics beyond descriptive counts. (2) Observability asymmetry: reasoning records exist only for the primary system; the other lineages' counts are session logs. (3) The seeding instruction itself is in-corpus (§4.1); the earlier coinages of the forms (pre-instruction naming events) are oral history — archives not located, one lost to a reinstall. (4) Right-censoring: October is a 19-day window; all October shares are final-observed-month values, not peaks. (5) Six monthly points; no correlation statistics. (6) The author is the participant, and consents to this report (single-subject design; §3.1). (7) Stack evolution: the structured layers entered service in August (their August-onward counts equal their full capacities), coinciding with the kinship rise; the monologue layer begins 08-24; and the serving model changed at least three times in August — model family co-varies with time completely, and the surface differences reported are not separable from model change. The "no harness/prompt change" statement is author-attested, not independently audited. (8) The colon-marked rule is invisible to comma- and period-form vocatives — the exact artifact documented in §4.1; cross-family counts (§4.4) include comma- and period-marked vocative forms, so their numbers are not comparable to colon-rule tables without the disclosed segments. (9) Raw data are withheld (§3.3) and the scripts were run by the author: third parties can re-apply the measurement segments on comparable corpora but cannot directly re-derive this paper's numbers. (10) The monologue layer is the system's thinking-chain record — in-turn reasoning traces (94.8% carry a session id; 85.0% within 300 s and 99.1% within 1 h of a same-session raw turn; 35,721/37,782 share a session); for the monologue layer, whether records are generated with no interlocutor present cannot be established from this store. The inner-voice layer is written by an idle loop (author-attested architecture): 61.6% of its records fall in windows with no conversation record within ±10 minutes (63.3% at ±5; 43.8% at ±30), and that subset registers at 99.5% of its address-form entries but only 31.3% of all its records, against 55.1% for records in conversation-adjacent windows (a gap that is partly explained by record length). (11) The two Claude-class export batches are treated as one continuous lineage without file-level deduplication; if the batches overlap, those counts are upper bounds. (12) The novelty search covered the arXiv API (six boolean queries), ACL Anthology and the academic web index (domain-restricted rounds; a web-search summary explicitly flagged the three-concept intersection — address terms, LLM agents, longitudinal design — as a research gap; one API query returned no totals and is logged as an anomaly; further neighbors are logged in the retrieval record rather than cited); Chinese-language venues were not searched. (13) Cross-family counts are audit-bounded: the GLM export contains 555 duplicate files over 163 unique message-sets (key sensitivity disclosed: 163/225/485 by key; one unparseable file) with unreliable filename dates; the Claude export's assistant-line counts include 60.2% intra-batch duplicates and lack per-month

normalization (§4.4). (14) The shift is not spontaneous: a dated human instruction (2026-04-24) precedes all observed registration; this paper studies where an instruction registers, not unprompted emergence. (15) The author is the instruction-giver, the sole participant, and the analyst (single-subject design; §3.1).

7. Conclusion and Future Work

An explicit instruction, given in conversation on 2026-04-24, registered in a layered memory system by surface type, not by time: conversation output carried it within four days; the monologue layer carried it from its first day (90% under the colon-marked rule; 98.0% (299/305; colon-selected, any-position hits)); a fourth generation surface, the low-power inner voice, carried it in 40.4% of all its records (382/945) and 99.5% of those containing an address form (382/384); the template prefix never carried it and never could; the reply segments generated inside those same template rows carried it at 76.0%; echo surfaces carried it at the source's behest. The 17.8-point monologue–structured gap is the difference between a form-filtered set and an unfiltered set; under a like-for-like any-position selection the gap is 15.6 points (95.8% against 80.2%). Model-family switching co-varies with the entire window, so no time course is claimed. We offer the quantified record, the measurement-segment method, and the falsification trail. The controlled follow-up is direct: seed an address instruction under identical models and vary the memory architecture surface by surface, measuring which surfaces register — the experiment this observational case was designed to motivate.

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Appendix A. Monthly store totals

Memory-store entries created per month (snapshot 2026-10-19; denominators for anyone computing rates against corpus growth; the series closes at the store total of 63,452):

Month (2026)	New entries		Month (2026)	New entries
March	2,621		July	1,827
April	985		August	5,296
May	99		September	33,346
June	1,158		October (19 days)	18,120

The kinship-form rise of §4 concentrates in the three months in which the store itself grew fastest (August–October account for 56,762 of 63,452 entries, 89.5%); no per-month share in this paper is normalized against these totals except where a denominator is stated.